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Department of CSE

B.Tech. CSE Trimester -5

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Cloud Infrastructure and Services

25CS2206E

LAB RECORD

2026

Experiment -1b

- **AIM: To create and configure Multi-Factor Authentication (MFA) in AWS using the AWS IAM Root User account.**
- **To improve AWS account security by adding an additional verification step during login.**

Description:

- MFA stands for Multi-Factor Authentication.
- It provides an extra layer of security to an AWS account.
- It requires a password and a second verification method to log in.
- The second verification method can be an authenticator app, passkey or security key.
- AWS supports applications like Google Authenticator and Microsoft Authenticator.
- MFA helps prevent unauthorized users from accessing the AWS account.
- It is recommended to enable MFA for the AWS Root User

Procedure to Create and Configure Multi-Factor Authentication (MFA) in AWS Using the Root User Account

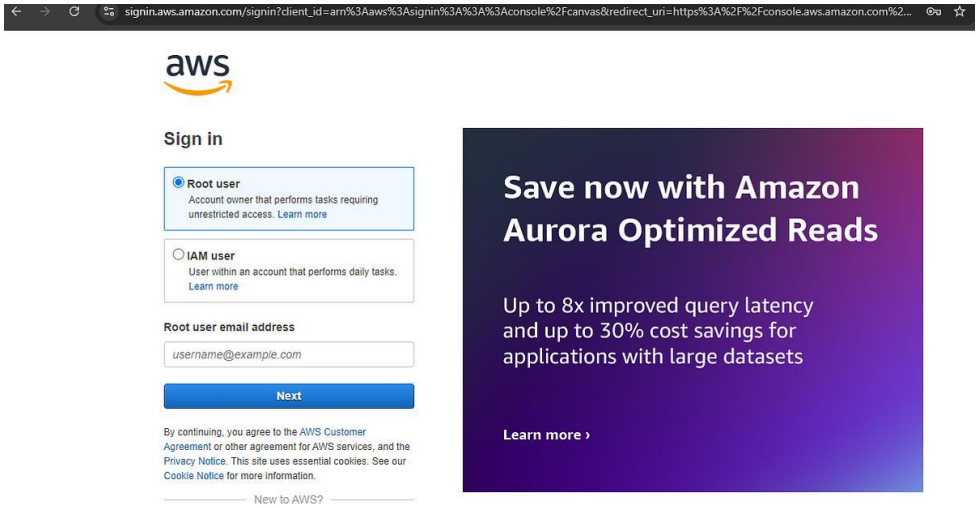
STEP 1: Open AWS Management Console and Log in

Explanation:

- The first step is to access the AWS Management Console using the Root User credentials.
- The Root User has complete access to the AWS account.

Navigation and actions:

1. Open Google Chrome on your computer.
2. Visit AWS Management Console.
3. Click Sign in to the Console.
4. Select Root user.
5. Enter your registered email address.
6. Click Next.
7. Enter your AWS account password.
8. Click Sign in.



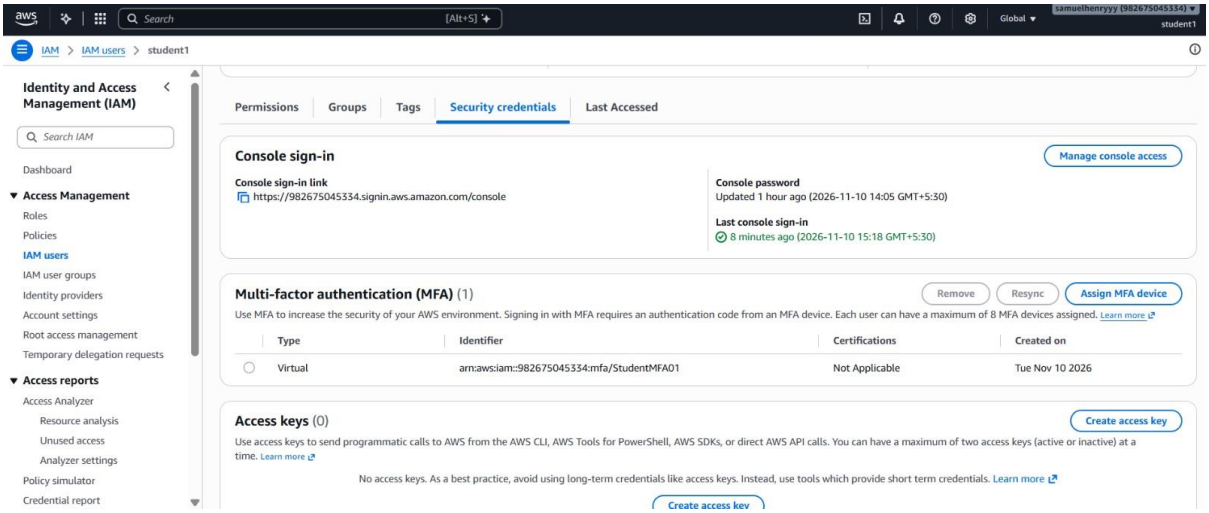
STEP 2 : Navigate to security credentials

Explanation:

- Security Credentials contains the security settings of the AWS account.
- It allows users to manage MFA devices and other security options.

Navigation and actions:

1. Look at the top-right corner of the AWS Console.
2. Click the account name or account menu.
3. Select Security credentials.
4. The security credentials page will open.



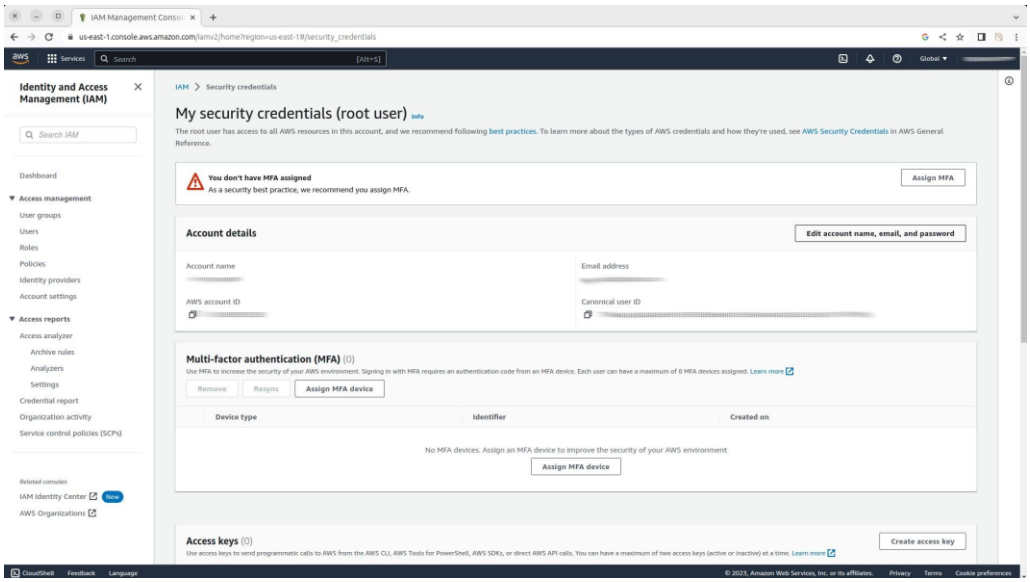
STEP 3 : Find the Multi-Factor Authentication (MFA) Option

Explanation:

- The MFA section allows the Root User to add an additional login verification method.
- It displays the MFA devices currently assigned to the account.

Navigation and actions:

1. On the Security Credentials page, scroll down.
2. Find the section named Multi-Factor Authentication (MFA).
3. Check whether any MFA device is already assigned.
4. Locate the Assign MFA device option.



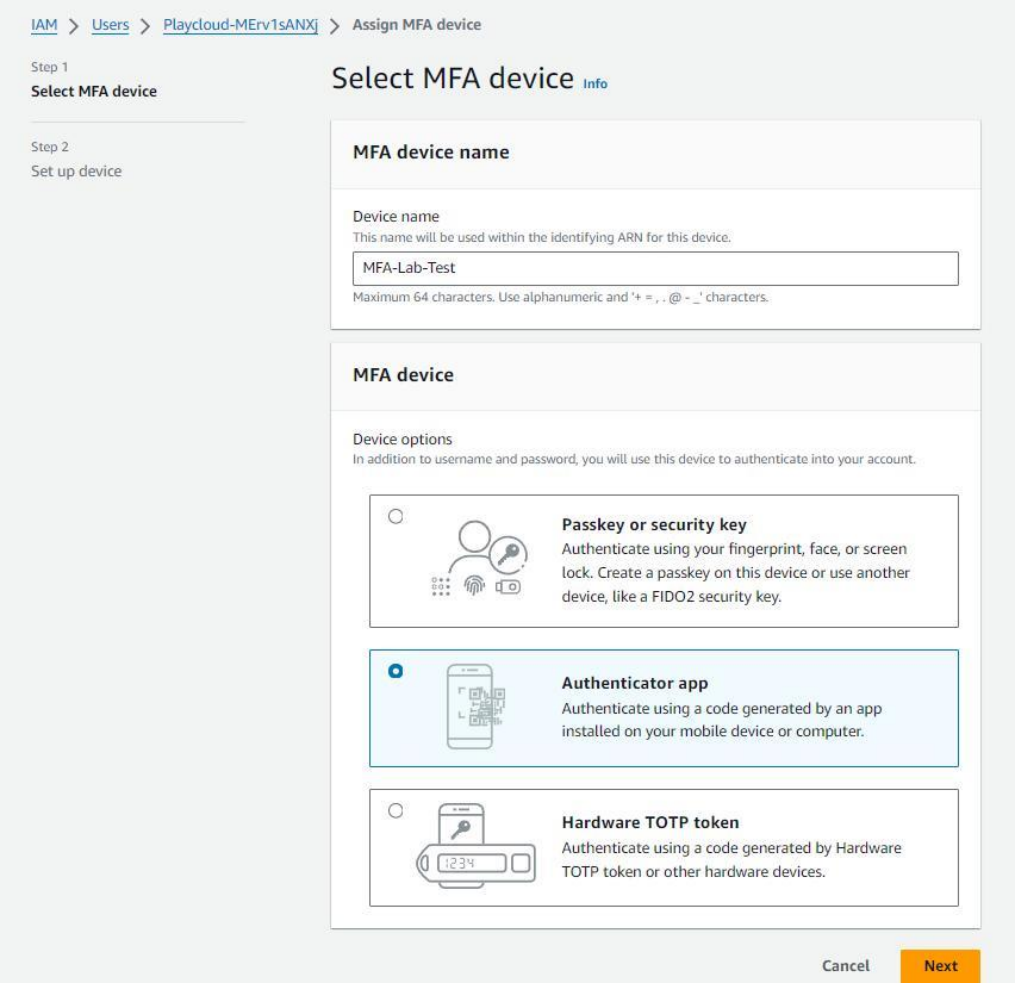
STEP 4 : Click on Assign MFA Device

Explanation:

- This option starts the process of registering a new MFA device with AWS.

Navigation and actions:

1. Click Assign MFA device.
2. A setup window will appear.
3. Enter a device name, for example, `RootMFA01`.
4. Select the appropriate MFA device type.
5. Choose Authenticator app.
6. Click Next.



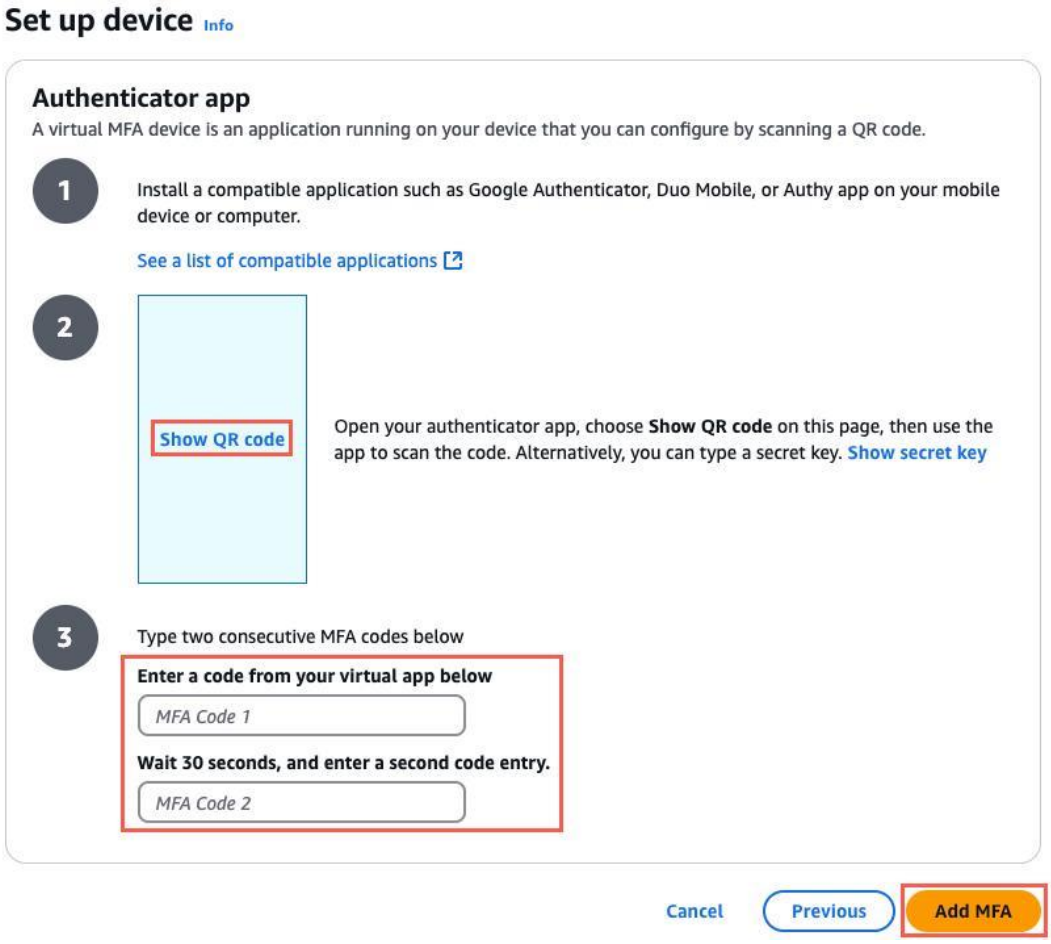
STEP 5 : Select an Authenticator App

Explanation:

- An authenticator app generates a temporary six-digit security code.
- AWS supports virtual MFA devices such as Google Authenticator and Microsoft Authenticator.

Navigation and actions:

1. In the MFA setup window, select Authenticator app.
2. Confirm the device name.
3. Click Next.
4. AWS will display instructions for connecting the authenticator app.



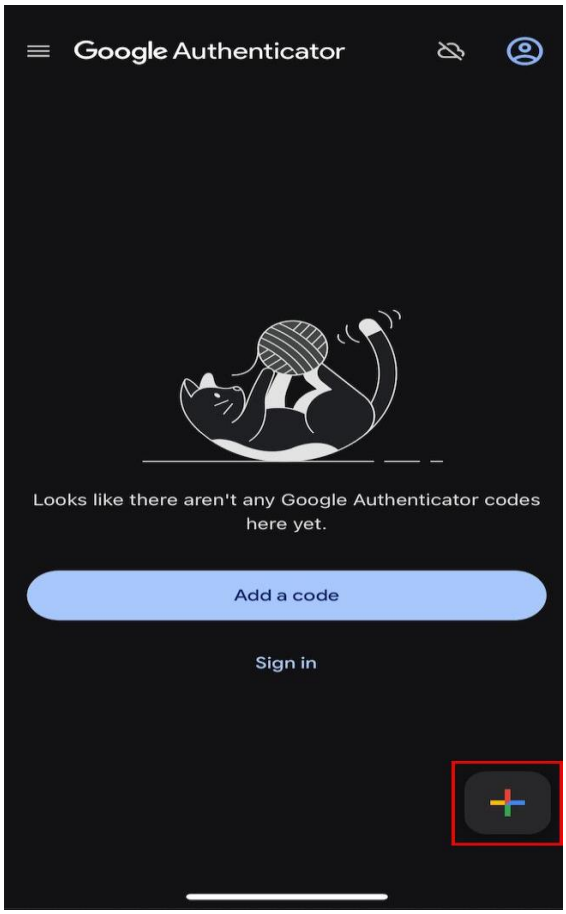
STEP 6 : INSTALL AND OPEN GOOGLE AUTHENTICATOR

Explanation:

- Google Authenticator is a mobile application that generates security codes.
- It is used to verify the identity of the AWS account owner.

Navigation and actions:

1. Take your smartphone.
2. Open Google Play Store (Android) or App Store (iPhone).
3. Search for Google Authenticator.
4. Select the official Google Authenticator application.
5. Tap Install or Get.
6. After installation, open the application.
7. Complete the initial setup if required.



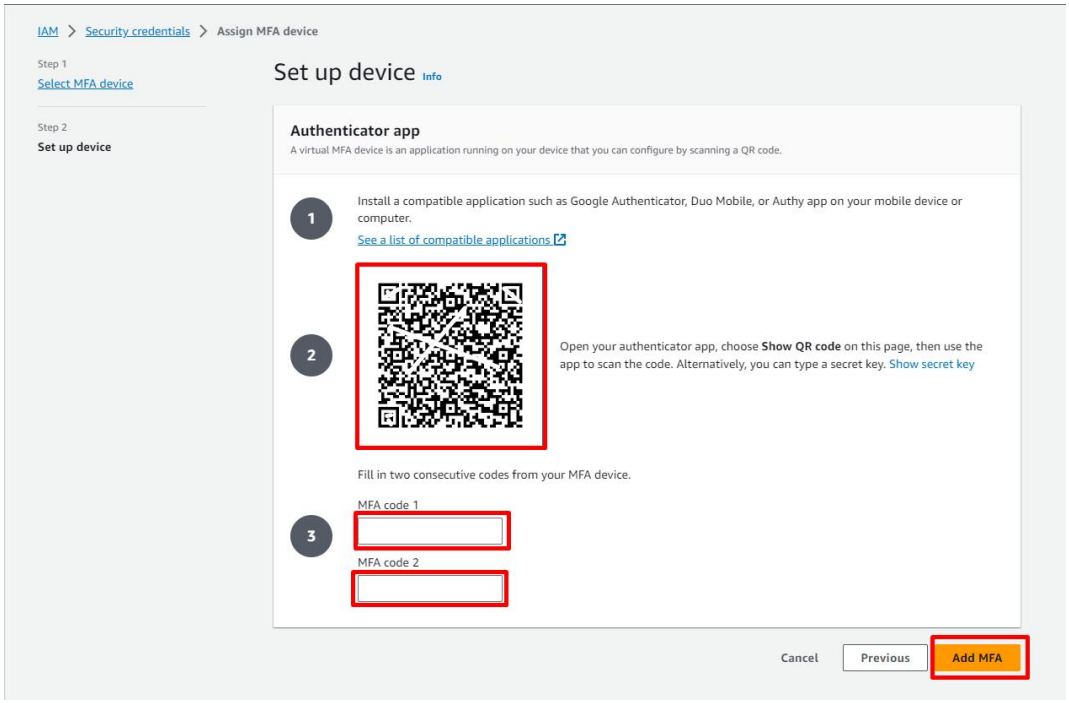
STEP 7 : Display the QR Code Provided by AWS

Explanation:

- AWS generates a unique QR code for the MFA device.
- This QR code contains the information required to connect AWS with Google Authenticator.

Navigation and actions:

1. Return to the AWS Console on your computer.
2. Open the MFA setup window if it is not already open.
3. Click Show QR code if the QR code is hidden.
4. Keep the QR code visible on your computer screen.
5. Do not share the QR code with anyone.



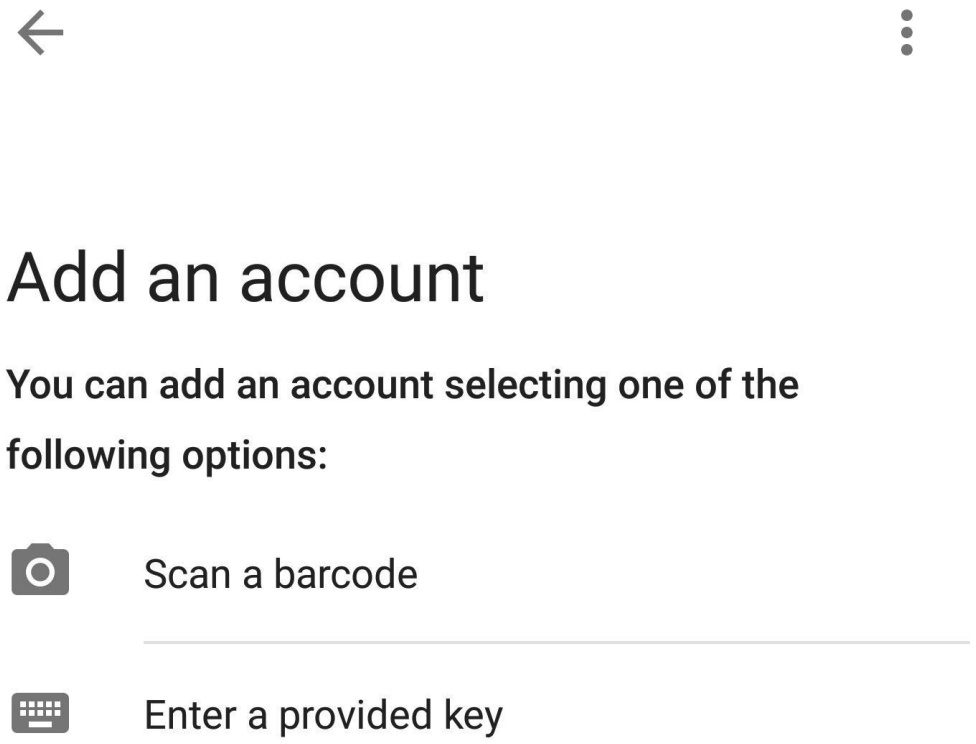
STEP 8 : Scan the QR Code Using Google Authenticator

Explanation:

- Scanning the QR code connects Google Authenticator with the AWS account.
- After scanning, the application starts generating six-digit verification codes.

Navigation and actions:

1. Open Google Authenticator on your mobile phone.
2. Tap the + (Plus) button.
3. Select Scan a QR code.
4. Allow camera permission if requested.
5. Point your phone camera towards the QR code displayed on your computer.
6. Wait for the application to scan the code.



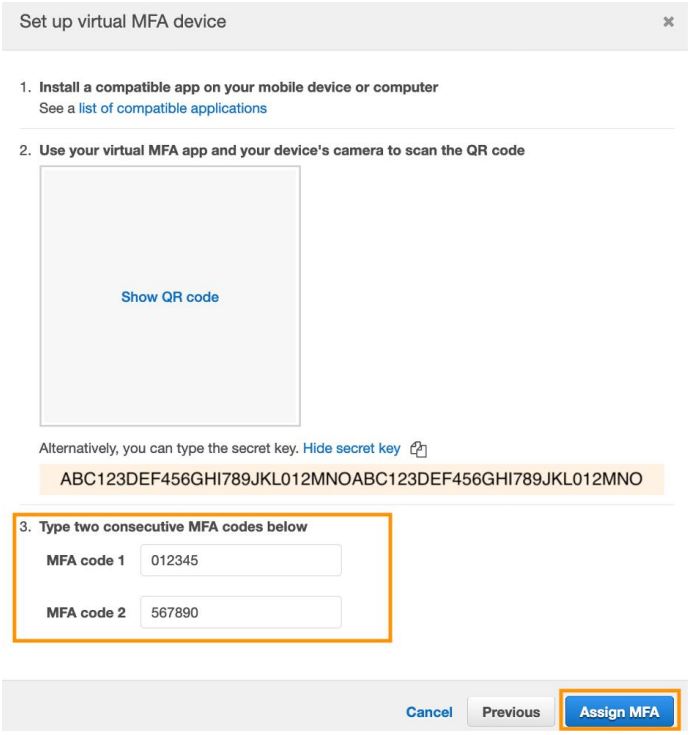
STEP 9 : Enter two consecutive MFA Codes

Explanation:

- AWS asks for two consecutive verification codes to confirm that the authenticator app is correctly connected.
- Both codes must be entered one after another.

Navigation and actions:

1. Look at the six-digit code in Google Authenticator.
2. Enter the first code in the MFA code 1 field in AWS.
3. Wait until Google Authenticator generates a new code.
4. Enter the new code in the MFA code 2 field.
5. Make sure both codes are entered correctly.



STEP 10 : Complete MFA Registration

Explanation:

- AWS checks the two verification codes.
- If the codes are correct, the authenticator app is registered as an MFA device.

Navigation and actions:

1. After entering both codes, click Add MFA or Assign MFA (depending on the screen).
2. Wait for AWS to process the request.
3. Check for the success confirmation.

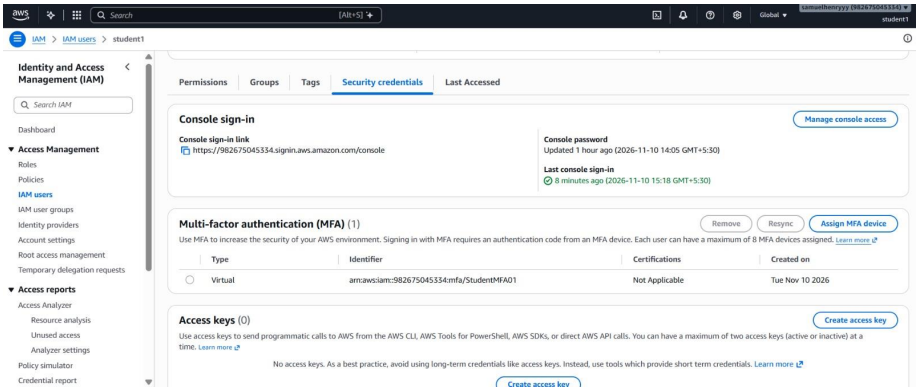
Step 11: Verify That MFA Is Successfully Enabled

Explanation:

- This step confirms that the MFA device has been registered successfully.

Navigation and actions:

1. Return to Security credentials.
2. Scroll down to Multi-Factor Authentication (MFA).
3. Check the list of assigned MFA devices.
4. Verify that the device name is displayed.
5. Check that the device type is Virtual.



Step 12: Log Out and Log In Again to Verify MFA Authentication

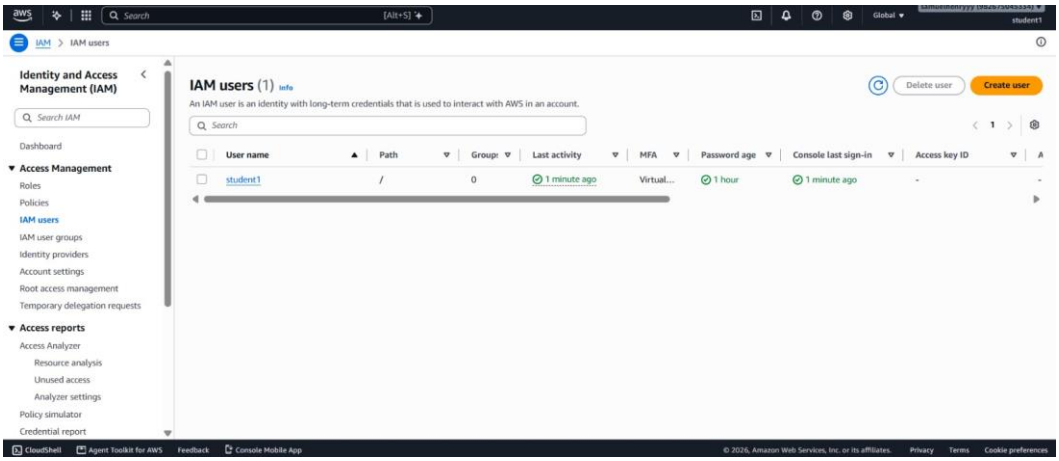
Explanation:

- This step checks whether AWS requests an MFA code during login.
- It confirms that the additional security feature is working correctly.

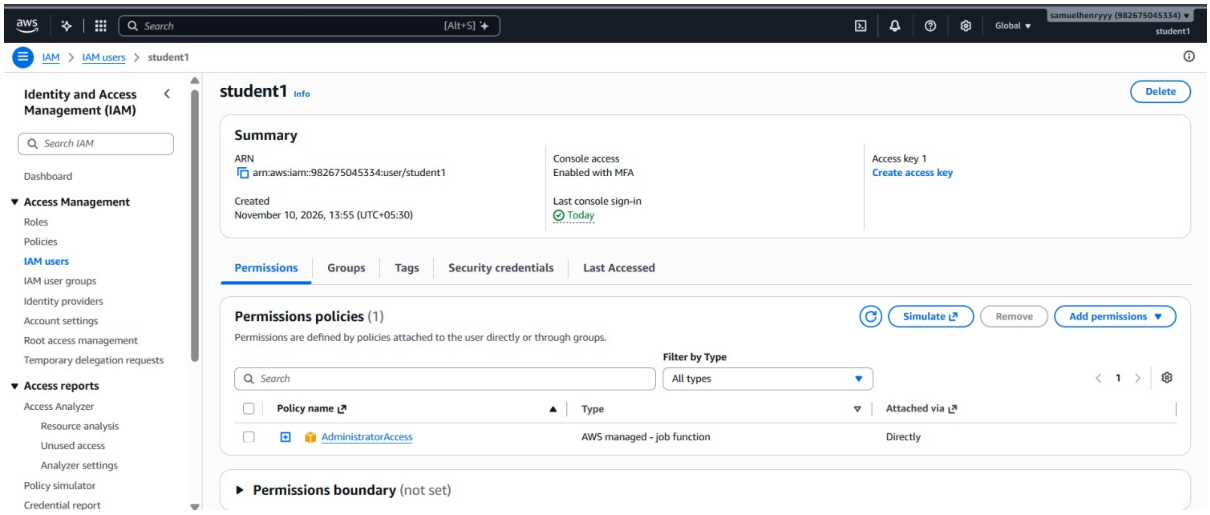
Navigation and actions:

1. Click the account menu in the top-right corner.
2. Click Sign out.
3. Return to the AWS login page.
4. Select Root user.
5. Enter the registered email address.
6. Enter the account password.
7. Click Sign in.
8. AWS will request an MFA verification code.
9. Open Google Authenticator on your smartphone.
10. Enter the current six-digit code.
11. Click Submit or Verify.

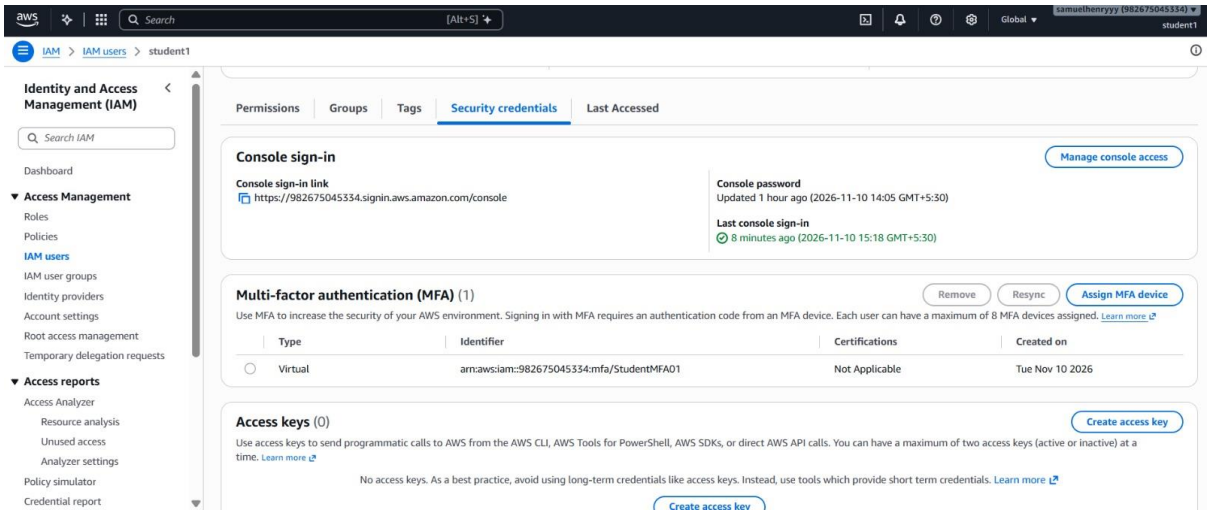
USER :

12. 

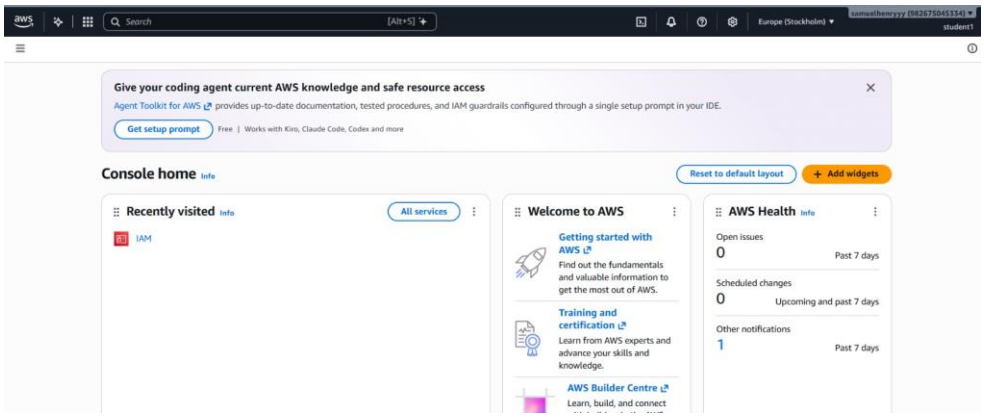
PERMISSION POLICIES



MFA SECURITY CREDENTIALS



CONSOLE HOME



RESULT

- Successfully enabled MFA for the AWS Root User account.

- Connected Google Authenticator with AWS.
- Verified MFA using a six-digit code.
- Successfully logged in with MFA authentication.
- Successfully created an IAM user named `student1`.
- Attached the `AdministratorAccess` permission policy.
- Configured a virtual MFA device using Google Authenticator.
- Verified successful login using MFA authentication.
- Confirmed that the user has the required permissions to access AWS services.
- Verified MFA authentication through successful login.
- Displayed the AWS Management Console homepage